

YouRule

Operation Guide

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1. Project Overview

1.1. What is YouRule?

YouRule is a real-time game management platform that allows non-technical users to design and run custom physical games without writing a single line of code. Using a visual, block-based editor powered by Blockly, users drag and connect blocks to define game rules, scoring conditions, zones, and events. YouRule then processes those blocks, converts them into executable logic, and referees the game automatically in real time.

The platform is built to be highly flexible. Users can create games in virtually any shape or format they imagine, from simple zone-based scoring to complex multi-step sequences involving passes, contacts, and timed events. It is suitable for anyone who wants to design and run a fun, structured physical game, whether that is a teacher running an activity in a classroom, a coach designing a training drill, or an individual creating a custom sport from scratch.

1.2. The Problem It Solves

Without YouRule, anyone running a physical activity game would have to manually track every movement and event happening in real time — who touched the ball, who entered a zone, who passed to whom — all without any assistance. This is error-prone, exhausting, and limits how complex a game can realistically be. YouRule automates this entirely: it tracks players and balls using Pozyx UWB positioning hardware, evaluates the game rules on every tick, detects fouls, awards points, and logs everything, acting as a fully automated digital referee.

1.3. Tech Stack at a Glance

Layer	Technology
Positioning hardware	Pozyx UWB (Creator One kit)
Backend	Python 3.11, Flask
Storage	Local JSON files
Frontend	Vanilla HTML, CSS, JavaScript

1.4. Design Philosophy

YouRule is built to run fully offline after initial setup. Once the hardware is configured and the software is installed, no internet connection is required. All processing happens locally on the host machine. This is a deliberate choice for reliability in environments where network access may be restricted or unavailable, and to ensure the system works consistently regardless of external connectivity.

2. Prerequisites

Before setting up YouRule, ensure the following hardware and software requirements are met.

2.1. Hardware - What You Need

YouRule uses **Pozyx UWB (Ultra-Wideband)** hardware to track where players and balls are located on the field in real time. UWB works like a very precise indoor GPS — small tags worn by players send signals to anchors placed around the field, and the system calculates each tag's position from those signals.

You will need the following physical components:

Anchors: Fixed devices placed around the edges of the playing field. They act as reference points for positioning. You need a minimum of 4 anchors for the system to work accurately. They should be placed at the corners of the field, elevated if possible.

Tags: Small devices carried by players or attached to balls. Each player gets one tag, and each ball gets one tag. Tags communicate with the anchors to report their position.

USB cable: One of the Pozyx tags (the master tag) connects to the host computer via USB. This is how the computer reads all the position data.

A Windows computer: YouRule runs on Windows. The computer does not need to be powerful, but it needs to be kept on and connected to the master tag for the entire duration of a game session.

2.2. Software - What to Install

You only need to install two things before running YouRule (note: you do not need to include the double quotation marks when typing commands):

Step 1: Install a modern browser

YouRule's interface runs in a browser. Use **Google Chrome** or **Microsoft Edge**.

Step 2: Install Python 3.11

Python is the programming language the YouRule server runs on. Download it from:

<https://www.python.org/downloads/>

Click the big yellow "Download Python Install Manager" button. Run the installer. On the first screen of the installer, make sure to tick "Add Python to PATH" before clicking Install.

To confirm if it is installed correctly, open the Start menu, search for **Command Prompt**, open it, type "python --version" and press Enter. You should see something like Python 3.11.x.

Step 3: Install Git

Download it from: [**https://git-scm.com/install/**](https://git-scm.com/install/)

Click "Download for Windows", run the installer, and leave all settings as default.

To confirm it installed, open Command Prompt and type "git --version" and press Enter. You should see something like git version 2.x.x.

Step 4: Clone the repository

Open Command Prompt, navigate to where you want the project to live (e.g. your Desktop) by typing "cd" followed by a space then type the path to your folder (e.g. Documents\YouRule Folder), and type "git clone <https://github.com/YouRule-App/YouRule-Game.git>".

This downloads the full project into a folder called "YouRule".

Note: if you cannot find the repository online, it means that it is private. Contact the admin of the system to give you permission.

3. Setup & Installation

Once Python is installed and the repository is cloned, follow these steps in order. You only need to do this once.

3.1. Setting Up the Virtual Environment

A virtual environment is an isolated space where YouRule's Python dependencies are installed, separate from anything else on your machine. Think of it as a self-contained box that holds everything the project needs to run.

Step 1: Open Command Prompt in the project folder. The easiest way:

- Open the “YouRule” folder in File Explorer (the folder where you previously cloned or placed the app in).
- Click the address bar at the top, type cmd, and press Enter
- A Command Prompt window opens already inside the right folder

Step 2: Create the virtual environment:

Inside the Command Prompt window, type “python -m venv venv”. This creates a venv folder inside the project. You only do this once.

Step 3: Test the virtual environment:

Inside the same Command Prompt window, type “venv\Scripts\activate.bat”. You'll know it worked when you see (venv) appear at the start of the command line, like this:
(venv) C:\Users\YourName\Folder\yourule>

3.2. Installing Dependencies

With the venv active, install the required Python packages by typing “pip install flask flask-cors pypoxyx”

This downloads and installs everything the server needs. It may take a minute. You only need to do this once — the packages are saved inside the venv folder.

To confirm everything installed correctly, type “pip list”. You should see Flask, flask-cors, and pypozyx in the list.

3.3. Connecting the Hardware

Before launching the system:

1. Place your 4 anchors around the field area — ideally at the corners, elevated off the ground if possible.
2. Connect the master tag to the computer through USB connection.
3. Make sure player and ball tags are powered on using a remote power source (e.g. a power bank).
4. Wait a few seconds for the computer to recognize the device.

3.4. Launching YouRule

Once hardware is connected and powered on, starting the system from this point on is always a single step, which is double-clicking on a file called “YouRule Launcher.bat” located inside the YouRule folder (where you cloned the project).

This automatically:

- Activates the virtual environment.
- Starts the server.
- Opens the Anchor Setup page in your default browser.

From the second launch onwards, the only steps you would need to do are step 3.3 and 3.4 when running YouRule. You can place the “YouRule Launcher.bat” file inside your desktop by right clicking on it then choosing the option “Create shortcut”.

4. Running the System

This section covers the day-to-day flow of using YouRule once everything is installed and set up. From this point on, running a game session follows the same sequence every time.

When YouRule Launcher.bat runs, two windows open on your computer and your browser launches:

"YouRule Server" window — a black console window showing the server logs. This is the backend running. You will see messages here as the system starts up and as games run. Keep this open for the entire session — closing it stops the server.

Launcher window — shows the message "Server is running. Close this window to stop." This window ties everything together. Leave it open until you are completely done with the session.

Browser — opens automatically to the Anchor Setup page. If it opens before the server is fully ready, wait 2–3 seconds and refresh.

You do not need to interact with the 2 windows that are automatically displayed on your screen whenever you launch YouRule. They are mainly used to report server status to developers.

4.1. Tag Address Mapping

The table below shows the correspondence between each tag's hex ID (as printed on the back of the small green boards) and its alphabetic ID, which is the identifier used by the system. Use this to check which tag you are referring to whenever you are trying to control or edit the tag's information on the browser.

Tag ID (printed on the back of the tag)	Tag ID (displayed on the web page)
0x6a62	A
0x6a2b	B
0x6a74	C
0x6a1f	D
0x6a01	E
0x6a5e	F
0x6a1e	Master tag (connected to your computer)

4.2. Building a Game

At launch, the Anchor Setup page is displayed for you to setup your anchor configuration. Once past the Anchor Setup page after clicking on the green “Continue” button, you land in the **Blockly Editor**. This is where games are designed.

To build a new game:

- Drag blocks from the left panel onto the workspace.
- Connect blocks to define players, balls, zones, rules, and actions (detailed explanation in section 5).
- Click **Save Game** to save your game to the server before running it.

To load an existing game:

- Click **Load Game**. A list of previously saved games appears.
- Click on one of the games to load it into the workspace.
- Review or edit if needed, then save your work and proceed to the Game Dashboard.

Unsaved changes will prompt a warning before you navigate away. Always save before running. By clicking the **Clear All** button located at the top left of the screen, you will delete all existing blocks in your current workspace, and the system will prompt you to confirm before doing this action.

To add or delete blocks:

- Click on one of the options on the drawer located on the left of your screen.
- Choose a block, then hold it and drag it to the workspace with your mouse.
- Hovering on a block will display a tip on how this block operates.
- Delete a block by clicking on it (the block will then be highlighted), then delete it by either pressing on backspace button on your keyboard or dragging it to the trashcan icon located at the bottom right of the screen.

4.3. Running a Game

From the Blockly editor, click **Run** to go to the Game Dashboard. From there:

1. Click **Load Game** to load the game you just built or imported
2. Review the score panel (left) — all players and their current scores
3. Review the field map (right) — live player and ball positions
4. Click **Start Game** when ready

During the game:

- The event log records every rule trigger, foul, throw, and timed event in real time.
- Scores update automatically as rules fire.
- The field map shows where every player and ball is at all times.
- The position sensitivity slider controls how smoothed the position readings are. Lower is faster but less precise, higher is smoother but slightly laggier

When the game is over, click **End Game**. Scores and the event log remain visible until you clear them or load a new game. A record of each game session with the players positions will be saved inside your project folder, under a sub-folder called “Records”.

When the session is finished:

1. Close the **launcher window** and the **server window** (the two windows that were initially displayed when the game was launched). This automatically shuts down the game server
2. Close the browser tab
3. Power off tags and anchors

4.4. Troubleshooting

Symptom	Likely cause	What to do
Browser opens but page doesn't load	Server still booting.	Wait 2–3 seconds and refresh.

Field map shows no dots	Positions not being processed by the server.	Check master tag USB connection, check if the tags are powered on, restart the server if the tags were powered back on.
Dots appear in wrong position	Anchor coordinates don't match physical placement, or the tag has been powered off then on while the game is running.	Update coordinates on the Anchor Setup page and save, restart the server if the tags were powered back on.
Game automatically ended	The game cannot read the tag positions.	Ensure the tags are powered on, then restart the server.

5. How the Game Works

5.1. Anchor Setup

Anchor Setup is always the starting point. Every launch begins here.

System Setup

Review your anchor configuration before starting. Press Continue when ready.

ANCHOR CONFIGURATION

ADDRESS	X (CM)	Y (CM)	
0x0c55	100	0	×
0x0c0a	100	100	×
0x6f3e	0	100	×
0x0c25	0	0	×

+ Add AnchorSave ChangesContinue

You can set how you want the anchors to be positioned by adjusting the x and y columns shown in the picture above. It is advisable to choose an anchor as your origin point (meaning x and y coordinates would be both zero) and then begin setting the other anchors' locations.

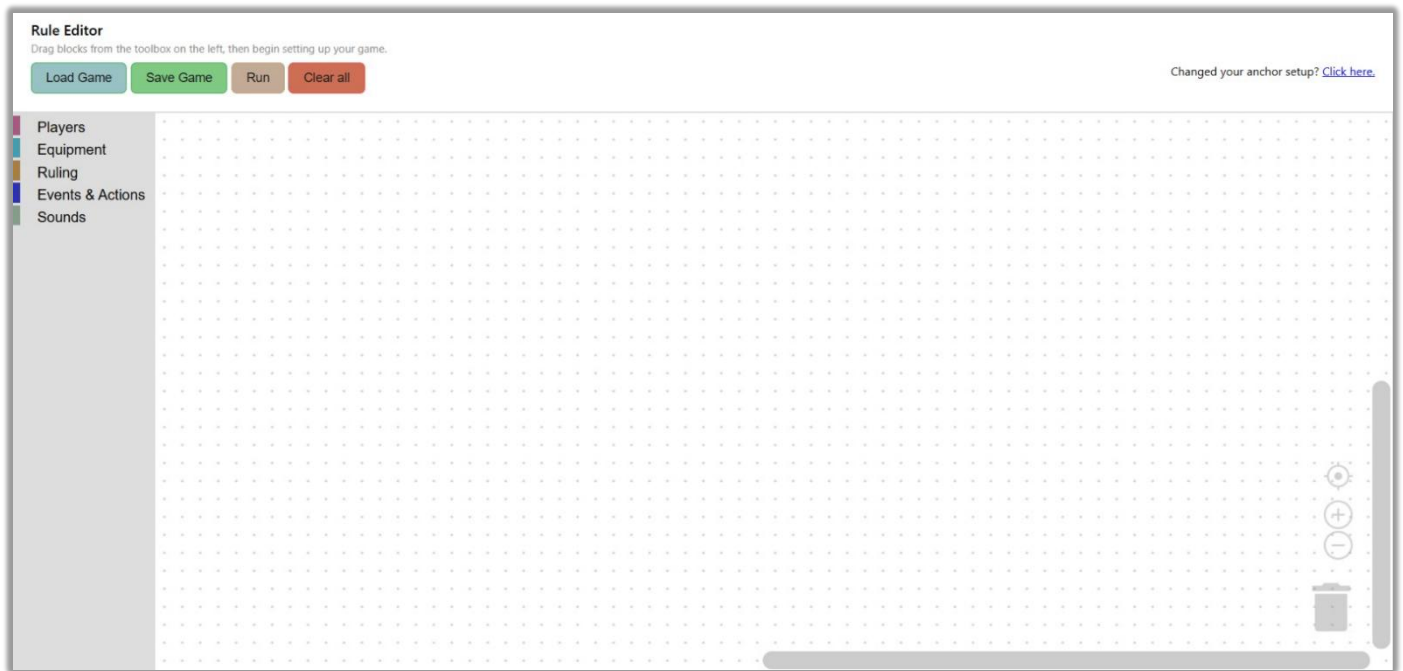
Once ready, the user can click on **Save Changes** if applicable and then click on **Continue**, which redirects them to the Rule Editor page.

5.2. Rule Editor

The Rule Editor section is the home base during game design. From here the user mainly does the following:

- Build and edit their ruling architecture and create a game environment.
- Manage tags, game zones, and sounds.

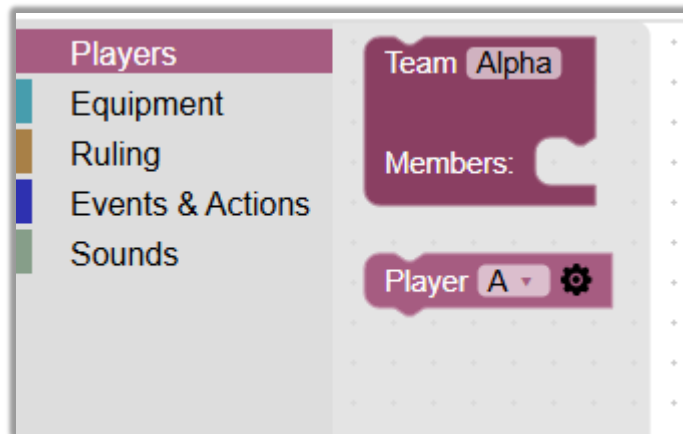
- Navigate to the main game dashboard upon finalizing their game rules.



On the left side of the screen, the user can drag out blocks from the drawer and start building their game using these blocks. There are 5 types of blocks the user can access:

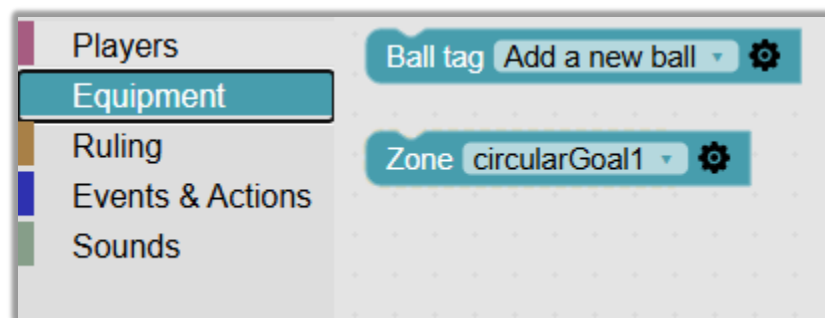
Players

- Team: represents a group of players that compete together under one name.
- Player: represents one individual player, identified by their Pozyx tag



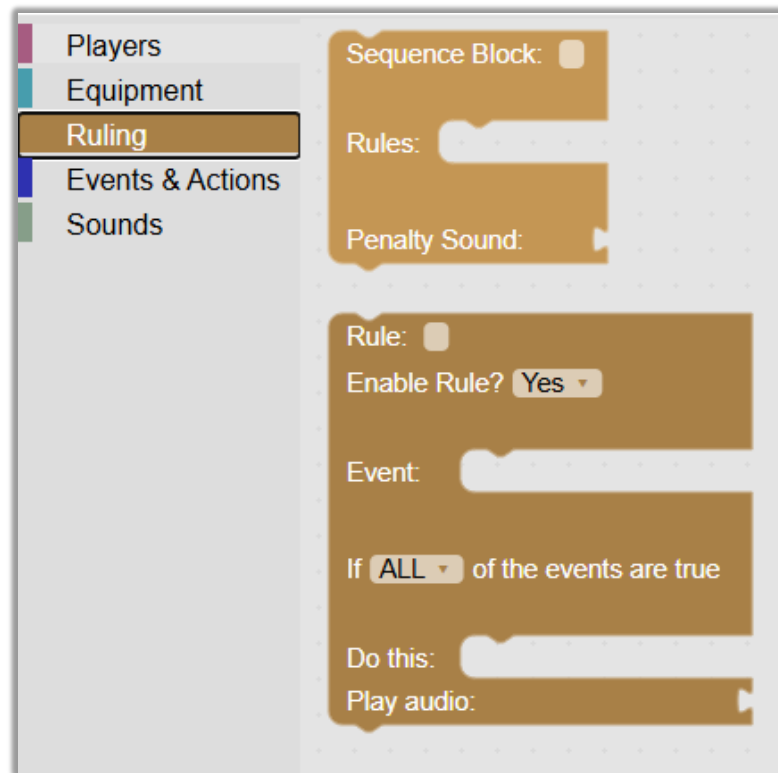
Equipment

- Ball: represents a physical ball being used in the game, identified by its Pozyx tag
- Zone: represents a named area on the field (e.g. a goal zone or boundary)



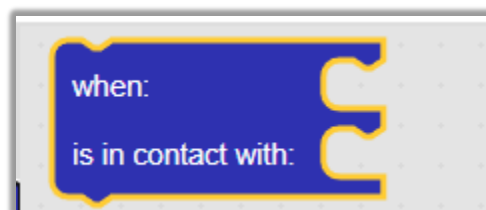
Ruling

- Sequence Block: a series of rules / events that must be completed in a specific order, one after another.
- Rule: a standalone rule that triggers whenever its conditions are met, regardless of order.



Events & Actions

- Contact: detects when two players are physically close enough to be considered in contact



- Ball Throw: detects when a specific player / team throws the ball to another specific player / team.



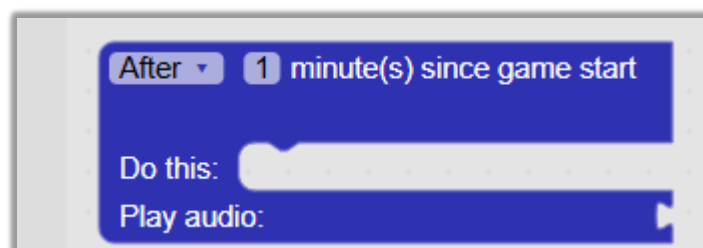
- Inside field: detects when a player or ball enters a specific zone on the field.



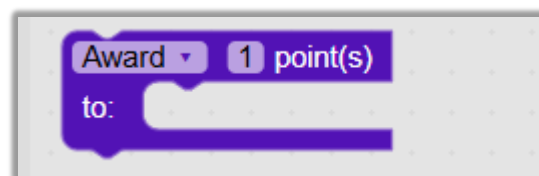
- Outside field: detects when a player or ball leaves or stays outside a specific zone.



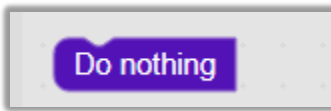
- Timed event: Triggers an action after a set number of minutes have passed since the game started.



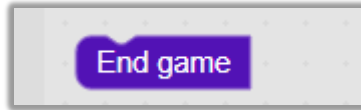
- Award: adds or subtracts a set number of points from a player or team when a rule fires.



- Nothing: used as a placeholder when a rule should trigger but no action is needed.

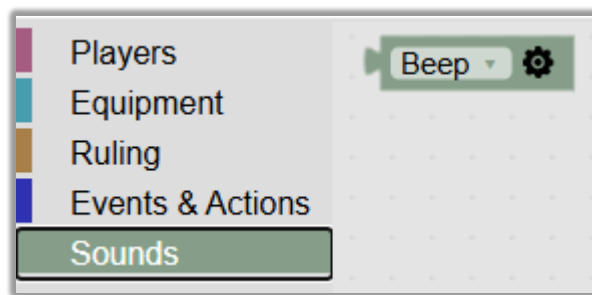


- End game: immediately ends the game when the rule it belongs to fires.



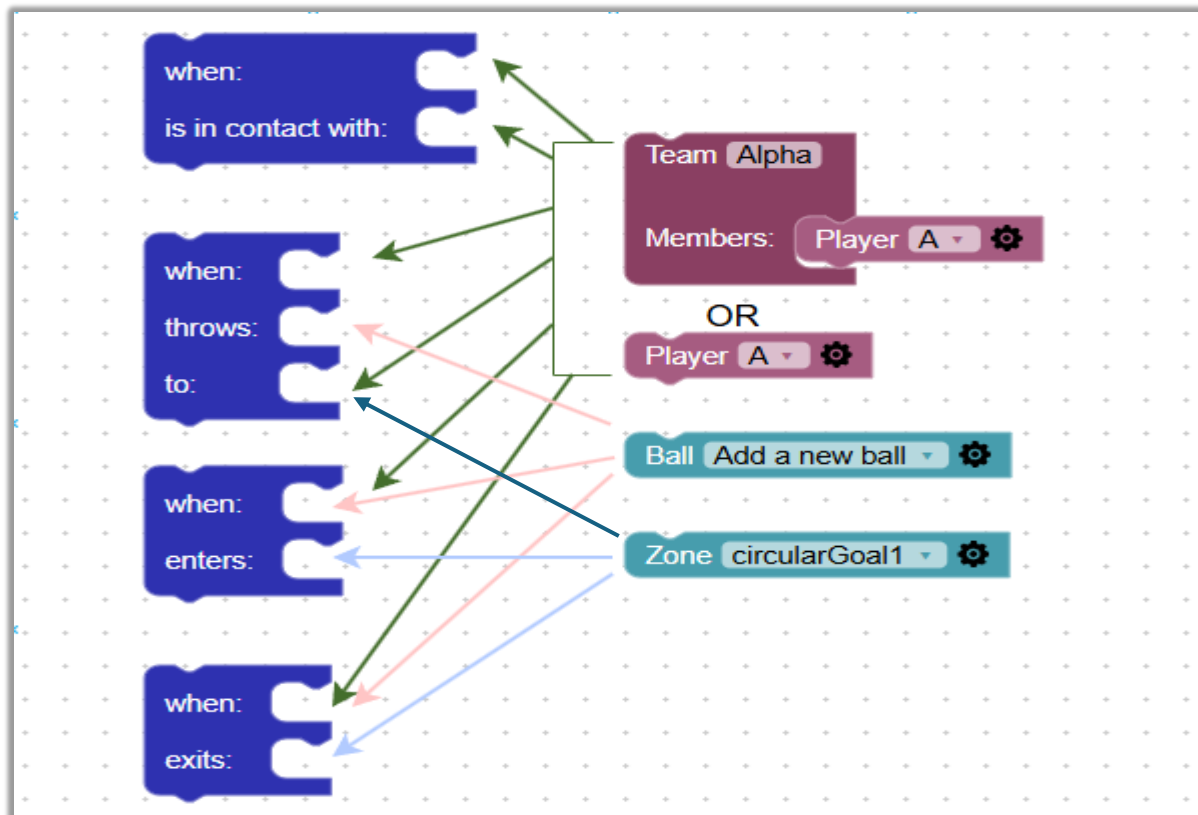
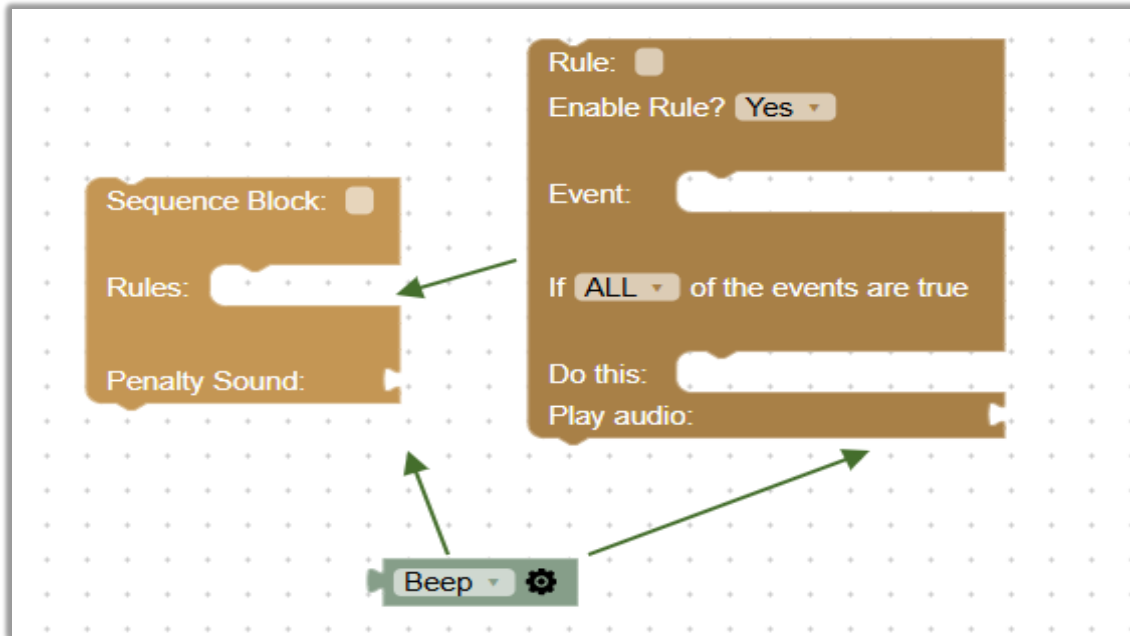
Sounds

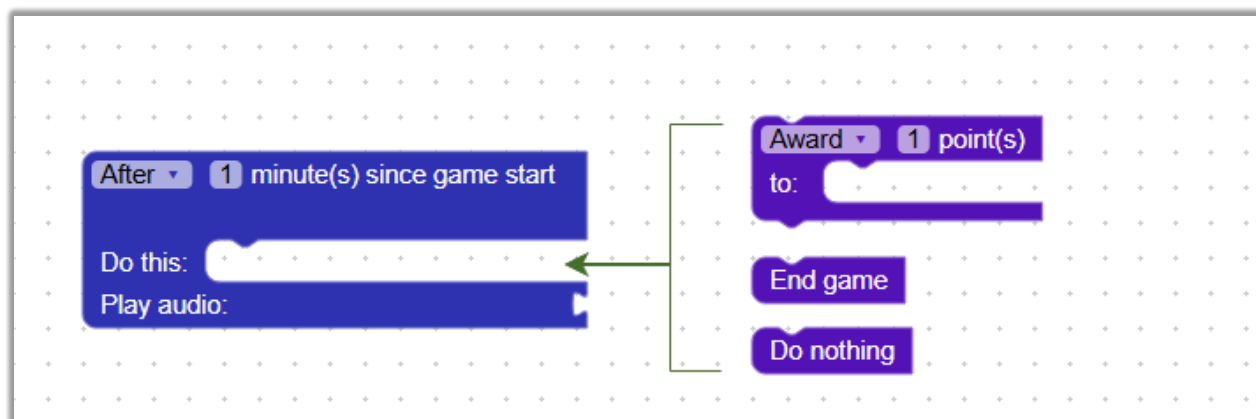
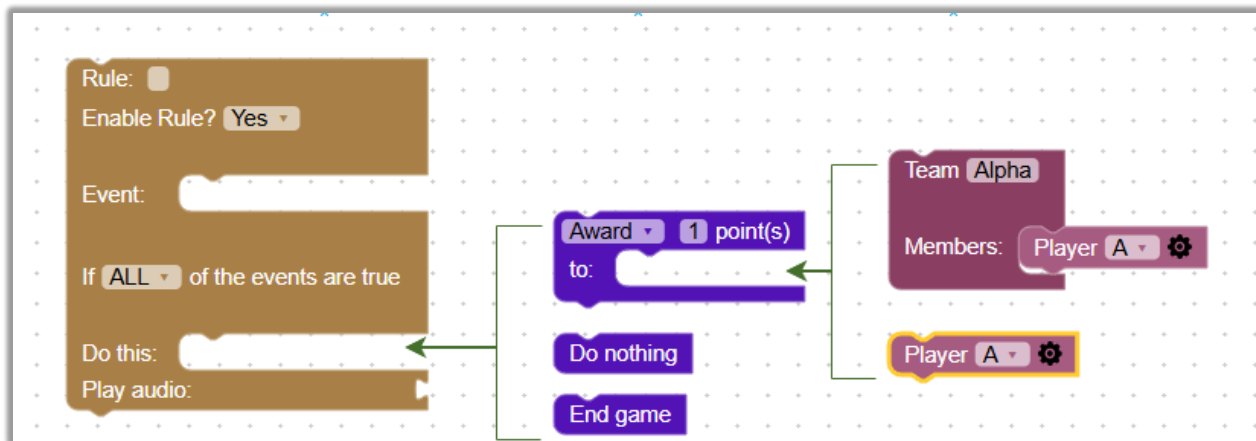
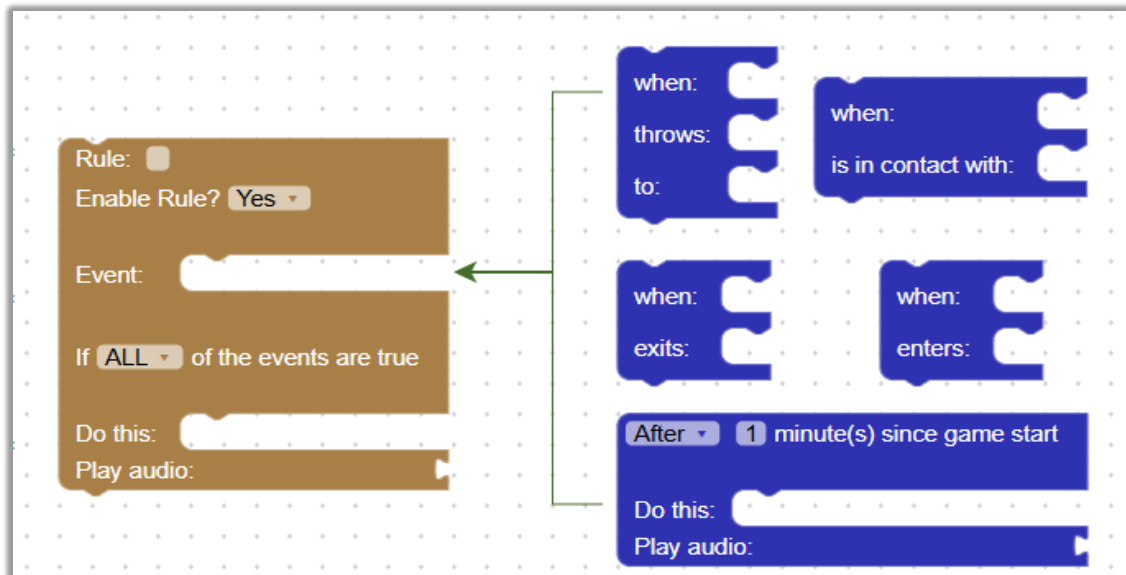
- Sound block: plays a chosen audio clip when the rule it belongs to fires



During rule setup, if you hover on a block, a text field will be shown that briefly explains what this block's function is, in case you get confused during rule setup. The "Enable Rule" field inside the rule block indicates whether this rule should be processed or not, decided by the user.

Each block that has input field accepts certain types of blocks. Figures shown below describe where each block goes





5.3. Customization and Zone Calibration

Some blocks contain a black gear icon as displayed in the previous images. These icons, when clicked, redirect you to a settings page. For instance, you can add / delete player and ball tags (where here add means activate and delete means deactivate), as well as set a custom name for each tag (which will update the block's dropdown menu with the custom name you have set). This can be something like the player's name or the ball type. Initially, the tags are named alphabetically.

Settings

Tags

Sounds

Zones

TAGS LIST

+ Add Tag

Modify tag names and roles (max character length: 10)

Tag A -

Player

▼



Tag B -

Player

▼



Tag C -

Player

▼



Tag D -

Player

▼



Save Tags

Clear Names

Back to Rule Editor

Back to Game page

Additionally, you can manage audios and zones, where you can change the name of the audio or zone previously saved, and add or delete new or existing audios or zones. You can also have a preview of the audio or zone by clicking on the speaker or gear icons beside their names respectively.

Settings

Tags

Sounds

Zones

AUDIO LIST

Modify audio clips names (max character length: 50) [+ Add audio](#)

Beep

Check

Clap

Goal

Whistle

Save Audios

Back to Rule Editor

Back to Game page

Settings

Tags

Sounds

Zones

ZONES LIST

Modify game zones names (max character length: 50) [+ Add zone](#)

circularGoal1

hexagon

Square

Withcirclezone

Withoutcirclezone

Save Zones

Back to Rule Editor

Back to Game page

Adding new zones involves going through a zone calibration process, where you are redirected to a page to start the process. You start by picking up any active tag that is currently used as a player or ball, then hold it and move towards the edges of the field, then drop the tag on the floor or surface and the system will record the tag's location relative to the anchor setup to create a zone that you can use to trigger events. There are two types of zones shapes you can choose from:

- Polygon: you set the number of corners that you would like to record, then you begin moving towards each corner as prompted, in a counterclockwise type of motion.
- Circular: you begin by moving to the center of the zone, then after recording that point you move towards the furthest points of the circle, which will be used to calculate the radius to form the circular-shaped zone.

Instructions are displayed at each stage of the calibration, and a progress bar is displayed to indicate the state of recording a point. At the end, you can see a preview of the zone that you have recorded, and you can manually adjust certain points recorded in case there was an offset observed during recording or if you want to change the shape geometrically (note that you cannot add / delete new or existing corners at the preview section, you will need to restart the calibration process by clicking on the “Restart” button at the bottom of the screen). Once ready, you can save your configuration by clicking on “Save Configuration” and give a name to the zone, which will be displayed as an option in the dropdown menu in the zone block, once you navigate back to the Rule Editor page.

Zone Setup & Calibration

1. Configure

Field Shape:

Polygon ▾

Calibration Tag Address:

A ▾

No. of corners to be recorded:

4

Start calibration

2. Capture points

Waiting to start...

0%

3. Save setup

No data yet.

Save Configuration

Restart

← Back to settings

In addition, adding new audios involves either manually recording a clip through the computer's microphone, or uploading an existing audio clip, then you give it a name then save.

Add a sound

Sound name

e.g. Whistle

Record from microphone

• Start recording

■ Stop

0.0 / 10.0s

— or —

Upload an audio file

Choose File

No file chosen

Save sound

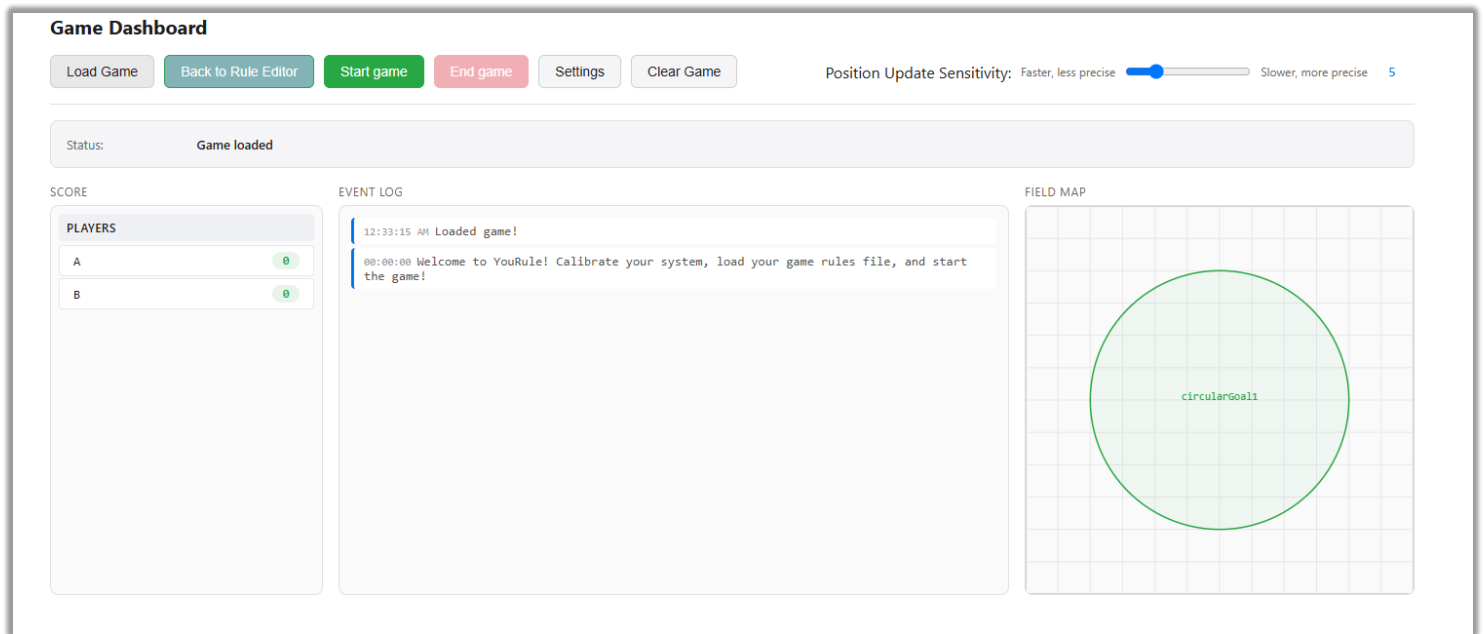
[← Back to settings](#)

5.4. Game Dashboard

Game Dashboard is where games are run. It is reached from the Blockly Editor via the “Run” button at the top of the screen. From the dashboard the user can:

- Import saved files from the Rule Editor and start the game.
- Monitor events displayed on the event log.
- Display the list of players / teams and their acquired points.
- Display the field map, with the zones inside it and the players’ and balls’ locations pinpointed on the map.
- Control the player and ball position update speed from the position sensitivity slider located at the top right of the screen.
- Navigate back to the Rule Editor or settings pages.
- End game.

In the background, the players' and optionally the balls' locations are continuously monitored, and rules / events are triggered primarily based on that (except for the Timed Event that is triggered based on time passed). At the end of the game, a file with the players' locations relative to the anchors' location is automatically saved locally inside a folder called "Records" located as a sub-folder inside the folder where you cloned / imported YouRule files.



5.5. Navigation Rules

A few things to be aware of when moving between pages:

Unsaved changes: if you navigate away from the Rule Editor without saving, a confirmation dialog warns you. The workspace is also auto-saved to session storage, so a refresh won't lose work, but navigating to a completely different page and back may not restore unsaved changes if the session was cleared.

Running game warning: clicking Back to Editor or Settings while a game is running shows a confirmation dialog. Proceeding ends the game. Scores and the event log are not saved automatically — note them before navigating away.

Server restart pages: any page that saves hardware configuration (anchor setup, add tag, settings tag changes) triggers a server restart. A loading overlay appears and the page stays put until the server is back. Do not navigate away during a restart.

Back buttons: every page has a back button that returns to the previous logical page. These are always in the bottom or top left of the page.